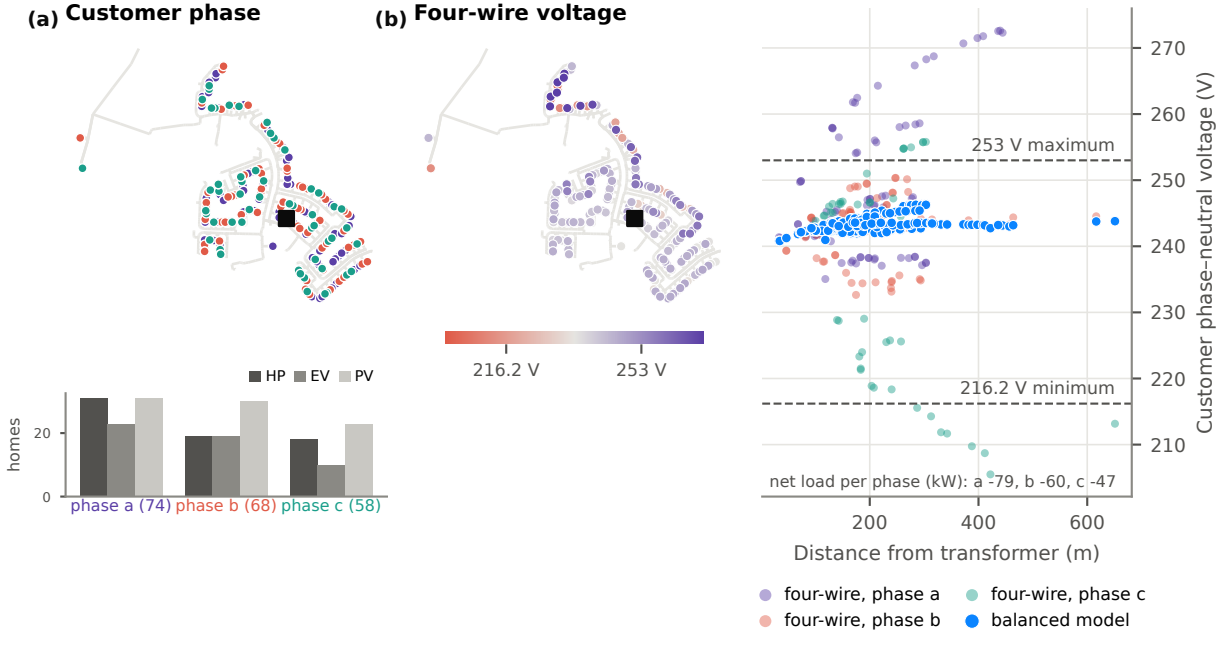

(a) Customer phase
(b) Four-wire voltage
(c) Same state along the feeder
HP
EV
PV
homes
20
0
phase a (74)
phase b (68)
phase c (58)
216.2 V
253 V
253 V maximum
216.2 V minimum
net load per phase (kW): a -79, b -60, c -47
Customer phase-neutral voltage (V)
270
260
250
240
230
220
210
200
400
600
Distance from transformer (m)
four-wire, phase a
four-wire, phase b
four-wire, phase c
balanced model